SUCHITRA K P

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PROFESSIONAL SUMMARY

Full-stack developer (React, Node.js, Express, MongoDB) with hands-on experience building multi-agent LLM systems using LangChain and applying ML to real-world engineering data. B.Tech, Electronics & Communication Engineering, PES University (CGPA 8.12/10).

EXPERIENCE

CHACS Lab, PES University — Machine Learning Intern *Jan 2026 – Apr 2026*

- Built an ML-driven pipeline to predict digital circuit power consumption, cutting reliance on repeated simulation runs by an estimated 40-50% for early-stage design analysis
- Benchmarked 4 predictive models (Decision Tree, Random Forest, Polynomial Regression, ANN), improving R^2 accuracy through systematic feature engineering, normalization, and hyperparameter tuning
- Reduced prediction error (MSE) across iterative tuning cycles, validating models on held-out test splits
- Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib

TECHNICAL SKILLS

- **Core CS Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), DBMS, Operating Systems, Computer Networks
- **Full Stack & Web:** React, Node.js, Express.js, HTML, CSS, JavaScript, REST APIs
- **AI / AI Agents:** LangChain (multi-agent orchestration), LLM integration, Prompt Engineering, Scikit-learn, ANN, Regression Models
- **Databases:** MongoDB, SQL
- **DevOps & Tools:** Docker, Jenkins, Git, GitHub, CI/CD Pipelines
- **Languages:** Python, Java, JavaScript
- **Other:** Cadence Virtuoso, Genus, SPICE (VLSI/EDA background)

PROJECTS

Multi-Agent AI Research System — LangChain

- Architected a 4-agent pipeline (Search, Reader, Writer, Critic) that autonomously researches a topic, extracts web content, and produces a structured, reviewed report end-to-end
- Designed shared-state handoff across all 4 pipeline stages, enabling coordinated multi-agent reasoning without manual intervention
- Built a Streamlit dashboard to run and visualize the full pipeline in real time

AI-Powered Chat Application with CI/CD

- Built a full-stack chat application with an automated CI/CD pipeline (Docker + Jenkins), reducing manual deployment steps and enabling consistent, repeatable releases
- Containerized the application stack for reproducible builds across environments

Other Projects

- **Airbnb Clone:** Full-stack booking platform with user authentication, REST APIs, and a persistent multi-user database layer
- **9T SRAM Compute-in-Memory for AES:** Designed digital modules for AES-128 encryption using a 9T SRAM-based CIM architecture in 45nm technology

COURSEWORK & CERTIFICATIONS

- **Relevant Coursework:** Machine Learning, Data Structures & Algorithms, Computer Architecture, Digital VLSI Design, CMOS VLSI Design
- Strong problem-solving foundation in Data Structures and Algorithms
- Continuous learner across Full Stack Development, AI Agent frameworks (LangChain), and Machine Learning